

ARTICLE INFO

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ABSTRACT

Here goes the abstract

1. Introduction

In recent years there has been a global expansion of renewable power generation Shahbaz, Siddiqui, Siddiqui, Jiao and Kautish (2023); Hassan, Algburi, Sameen, Al-Musawi, Al-Jiboory, Salman, Ali and Jaszczur (2024), driven by de-carbonization efforts and declining costs in renewable technologies Jäger-Waldau (2025); Joshi, Rao and Banerjee (2025); Manservigi, Castorino, Spina, Venturini and Gonzalez-Salazar (2026). This rapid deployment of renewable generation has been mostly due to additions in solar (at both utility and residential scale), hydro and wind capacity Shahbaz et al. (2023); Hassan et al. (2024). Currently, annual renewable capacity additions outpace conventional capacity additions IRENA (2025), and this upward trend will likely continue in the forecoming decades Agency (2019); Raturi (2021).

However, the rapid expansion of renewable generation is leading to grid integration difficulties due to inherent variability and uncertainty in the output Li, Shi, Cao, Wang, Kuang, Tan and Wei (2015); Liu and He (2023). Spatiotemporal mismatches between power supply and demand may arise if capacity planning and grid infrastructure development do not keep pace Ming, Kun and Jun (2013); Bird, Lew, Milligan, Carlini, Estanqueiro, Flynn, Gomez-Lazaro, Holttinen, Menemenlis, Orths et al. (2016); O'Shaughnessy, Cruce and Xu (2020); Li, Wang, Guo, Xin, Zhang, Liu and Feng (2024). Oversupply of renewable generation — occurring during periods of low demand or high wind and solar resource availability — can result in curtailment of a fraction of the generated energy Denholm, O'Connell, Brinkman and Jorgenson (2015); Bird et al. (2016); O'Shaughnessy et al. (2020). This issue is further compounded by the increasing penetration of residential-scale distributed generation, which is generally excluded from centralized dispatch scheduling owing to its negligible individual capacity Li et al. (2015); Bhattarai, Maraseni and Apan (2022). As a result, these challenges place increasing demands on power system flexibility and energy storage capacity, both of which are essential for balancing supply and demand in high-penetration renewable

energy systems Denholm and Mai (2019); ?; Nejla, Saidani and Besbes (2025).

In particular, South America has also experienced accelerated penetration of renewable generation in recent years, with wind and solar photovoltaic capacity accounting for more than half of annual capacity additions International Energy Agency (2023). Although renewable penetration has been uneven across countries, several are already experiencing increasing curtailment rates von Papp, Moraga and Henríquez (2022); Silva, Renolphi, Vieira, Lourenço, Salles and Monaro (2025). Chile has recorded curtailment rates of up to 20% of total annual renewable generation, driven mainly by a geographical mismatch between solar generation — concentrated in the northern Atacama region — and demand centers located farther south von Papp et al. (2022); Silva et al. (2025). In Brazil, by contrast, the rapid expansion of distributed generation has reshaped the load profile, also resulting in high curtailment rates ?. Despite these challenges, the region's abundant hydroelectric capacity ?International Energy Agency (2023) represents an additional advantage for renewable energy integration, offering flexible storage and dispatch capabilities that can help mitigate variability and curtailment ?.

Bolivia's current power mix is dominated by hydro and thermal generation—accounting for approximately 70% and 20% of installed capacity, respectively — with only a small share of non-hydro renewables ? —. Efforts are underway to further decarbonize the electricity sector: national development plans ? and Bolivia's Nationally Determined Contributions (NDC) ? foresee substantial growth in centralized renewable capacity. The accelerated uptake of distributed solar PV observed in the region ? suggests that distributed generation growth represents a plausible complementary pathway.

Given this growing complexity in power systems, energy modeling has become a crucial tool for exploring supply and demand scenarios that support long-term policy decisions ?. However, few studies have modeled the impacts of renewable energy penetration on power systems in South America. In Bolivia, Fernandez-Vazquez et al. ? developed a long-term optimization model of the Bolivian energy sector using OSeMOSYS, based on a trend-based projection of power demand, and analyzed policy-based scenarios to induce decarbonization of the power system. However, the

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specific impacts of renewable capacity expansion — particularly distributed generation — on transmission requirements, thermal fuel consumption, greenhouse gas (GHG) emissions, and energy curtailment remain unexplored. This study aims to analyze the impacts of exogenous renewable capacity additions — both centralized and distributed — on power transmission, fuel consumption in thermal power facilities, GHG emissions, and energy curtailment. An intraday hydroelectric dispatch strategy is also proposed to mitigate renewable energy curtailment. For this purpose, we constructed a regionalized model of the Bolivian energy sector using the LEAP-NEMO tool.

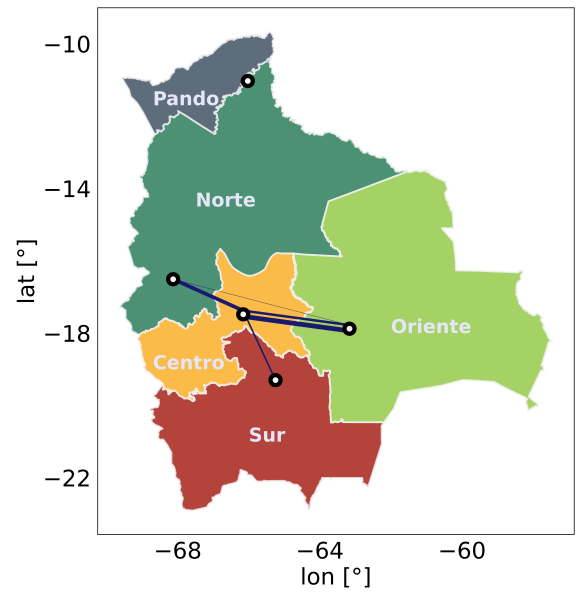


Figure 1: Regionalization included in the model and power lines

Figure 2:

Table 1

2.

CRediT authorship contribution statement

Emilio Bianchi: Conceptualization of this study, Methodology, Software. **Ignacio Sagardoy:** Conceptualization of this study, Methodology, Software. **Francisco Lallana:** Conceptualization of this study, Methodology, Software. **Gustavo Nadal:** Conceptualization of this study, Methodology, Software. **Gonzalo Bravo:** Conceptualization of this study, Methodology, Software. **Gonzalo Bravo:** Conceptualization of this study, Methodology, Software.

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